Executive Summary: Geometric Foundations for Unified Physics: Empirical Consequences and Experimental Tests

1. Purpose & Motivation

While general relativity (GR) and quantum mechanics (QM) each describe aspects of our universe with remarkable precision, gaps remain—especially in regimes where strong gravity meets quantum phenomena (black holes, early universe, vacuum fluctuations). Many efforts of unification emphasize theoretical beauty but seem to struggle in delivering concrete, testable predictions. This framework seeks to bridge the gap by focusing on falsifiable consequences with current and near-future experiments.

2. Framework in Brief

This model introduces three emergent structural constraints—Zero (collapse), Infinity (expansion), and Chance (probability)—which operate alongside space and time as "dimensions" embedded directly in the metric. We don't invoke new particles or exotic forces - the constraints provide clear geometric boundaries for collapse, expansion, and quantum uncertainty. The aim is a minimal, testable extension: everything added yields a direct, observable consequence.

3. Experimental Signatures

- Gravitational Waves:

The framework predicts subtle, but measurable deviations in polarization states and waveform propagation—particularly in strong-field, merger, or high-frequency regimes. These signatures may manifest as anomalies in LIGO, Virgo, KAGRA, or future LISA data.

- Black Hole Observables:

Modifications to the metric near singularities may influence black hole shadow geometry, photon ring structure, and horizon-scale emissions—potentially visible to the Event Horizon Telescope or next-gen observatories.

- Quantum Interference and Vacuum Fluctuations:

The "Chance" dimension is expected to produce slight, geometry-driven departures in phase noise, decoherence rates, and zero-point fluctuations—potentially testable in high-precision atomic clocks, quantum optics, or Casimir-type setups.

- Cosmic Microwave Background (CMB) and Large-Scale Structure:

Here, the framework predicts distinctive imprints on polarization patterns, non-Gaussianity, and small-scale anisotropies in the CMB - Also possible signatures in cosmological surveys (e.g., CMB-S4, Euclid).

4. Paths to Measurements

These effects come with experiment in mind—not as speculative signatures forever out of reach, but as targets for ongoing or imminent investigations:

- Gravitational wave observatories: LIGO, Virgo, KAGRA, LISA.
- Black hole imaging: Event Horizon Telescope, ngEHT.
- Quantum metrology: atomic interferometers, squeezed-light setups, quantum optomechanics.
- Cosmology: CMB-S4, Planck data reanalysis, galaxy surveys.

If these signatures are not found within present experimental sensitivity, the framework is directly falsified. Something not feared, but beyond full reach of the author.

5. Falsifiability & Invitation

A core commitment of this model is explicit, empirical falsifiability. The outlined deviations—polarization anomalies, black hole shadow geometry, phase noise, CMB features—are all subject to concrete measurement or exclusion by near-term data.

I welcome collaborative critique, experimental proposals, or joint analysis. If any outlined prediction connects with your ongoing work, I am happy to provide full mathematical details, derivations, or work with your team to explore and learn about practical measurement strategies.

6. Relation to Ongoing Work

This framework is designed to interface with the experimental frontiers of gravitational wave science, quantum measurement, and black hole physics—fields where I understand your research has led, and could continue to lead, to major advances. Unlike many "theory-heavy" models, all new structure here is added for its testable consequence, not for abstraction.

7. Conclusion & Contact

Thank you for considering these ideas. I think any true unification hypothesis must be subject to experimental challenge—not just theoretical elegance.

Full derivations, appendices, and the complete paper are available on request.

Contact:
Jed Kircher
916-396-8071 (U.S.), +420 728 77 44 94 (CZ)
Jed.kircher@gmail.com
Sacramento, CA / Prague